DOUBLY ISOLATED BATYREV MIRROR PAIRS AND NON-SMOOTHABLE CALABI–YAU THREEFOLDS

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Abstract. For a reflexive 4-polytope Δ with unit edges the generic anticanonical hypersurface $X \subset \mathbb{P}_\Delta$ is a Calabi–Yau threefold with isolated singularities; we ask when the Batyrev mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ is isolated-singular as well (we then call the pair *doubly isolated*). An exact identity (the lattice length of the dual edge of a two-dimensional face equals the multiplicity of its transverse cone) shows this happens if and only if every two-dimensional face of Δ is transversally smooth; we call such Δ *both-sides unit*. We determine the both-sides unit polytopes completely, subject to the database-copy completeness and no-duplication hypothesis stated in §5. Under that hypothesis an exhaustive scan with exact integer predicates and exact re-verification finds that, among the 473,800,776 reflexive 4-polytopes of the Kreuzer–Skarke classification there are exactly 590, and every two-dimensional face of every one of them is a triangle, a zonotope, or a reflexive polygon (one interior point). Consequently the non-smoothable germs occurring on doubly isolated mirror pairs are exactly two cyclic quotient points, $\frac{1}{3}(1,1,1)$ and $\frac{1}{5}(1,1,3)$, and the anticanonical cone over $\mathbb{F}_1$. The germs that deform but admit no smoothing (type (D) in the Altmann–Corti–Filip–Petracci trichotomy, the smallest being the cone over the Minkowski sum of a triangle and a unit segment) never occur: a Calabi–Yau threefold with isolated singularities carrying one never has a mirror with isolated singularities. At the opposite extreme, exactly one reflexive 4-polytope yields a mirror pair with both members *smooth*: the self-dual 24-cell, whose hypersurface is the self-mirror threefold with Hodge numbers $(20,20)$. The $\mathbb{F}_1$-cone occurs exactly once: a unique pair of polytopes, with 22 and 26 vertices, whose hypersurfaces form a mirror pair, with Hodge numbers $(20,26)$ and $(26,20)$ for the MPCP resolutions, in which X is non-smoothable while every singular point of X° is locally smoothable. We also record, as a further obstruction, that the 26 node-diagonal vectors of X° admit no all-nonzero relation: any global smoothing would have to couple the nodal directions to the four del Pezzo-cone directions. Finally, as a caution, the natural conjecture suggested by the low-vertex data (that only triangles and zonotopes occur) is true for all 395,406,329 polytopes with at most 18 vertices and false from 19 vertices on.

1. Introduction

Let $\Delta \subset N_\mathbb{R} = \mathbb{R}^4$ be a reflexive polytope all of whose edges have lattice length one, let $\Delta^\circ \subset M_\mathbb{R}$ be its polar dual, and let $X \in |-K_{\mathbb{P}_\Delta}|$ be a generic anticanonical hypersurface in the Gorenstein Fano toric fourfold associated to the face fan of Δ. (Throughout, Δ is the *fan* polytope, Batyrev's Δ^*, so our Δ° is his Δ; the same convention is in force in [Wue26a, Wue26b].) By Batyrev [Bat94], X is a Calabi–Yau threefold with Gorenstein canonical singularities, and by [Wue26a, Prop. 3.1] its singular locus is exactly: for each two-dimensional face $F \preceq \Delta$ whose cone $C(F)$ is singular, $\ell(F^\circ)$ points at which the germ of X is analytically isomorphic to $C(F)$, where $\ell(F^\circ)$ is the lattice length of the edge $F^\circ \preceq \Delta^\circ$ dual to F. The Altmann–Corti–Filip–Petracci trichotomy, recalled as [Wue26a, Thm. 2.2],

Date: August 2, 2026.

2020 *Mathematics Subject Classification.* 14J32 (primary); 14J33, 14M25, 14B07, 52B20 (secondary).

Key words and phrases. Mirror symmetry, Batyrev duality, reflexive polytopes, Kreuzer–Skarke classification, Calabi–Yau threefolds, isolated singularities, smoothability.

classifies $C(F)$ by the combinatorics of F into rigid (R), deformable-but-non-smoothable (D), and smoothable (S); a single face of the first two types forces X to admit no smoothing, and [Wue26b] studies when the type-(S) configurations smooth.

Mirror symmetry, in the Batyrev picture, is the polar duality $\Delta \leftrightarrow \Delta^\circ$, which for 4-polytopes exchanges two-dimensional faces with edges. This note asks the mirror question: when are X and the mirror $X^\circ \subset \mathbb{P}_{\Delta^\circ}$ *both* isolated-singular (we then call the pair *doubly isolated*), and which local models, in particular which non-smoothable ones, can occur in that situation?

The entry point is exact and elementary. For a two-dimensional face F write $u_1, u_2 \in M$ for the primitive inner normals of the two facets of Δ containing F, so $F^\circ = \operatorname{conv}\{u_1, u_2\}$, and let $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$. We call the two-dimensional cone $\operatorname{Cone}(u_1, u_2)$ the *normal cone* of F and its dual the *transverse cone* of Δ along F (the local model of Δ in the directions normal to F), and we say Δ is *transversally smooth* along F when this cone is smooth.

Theorem 1.1 (= Lemma 3.1 with Remark 3.2)**.** *$\ell(F^\circ) = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2\,]$, the multiplicity of the normal cone* $\operatorname{Cone}(u_1, u_2)$ *of F (equivalently, of its dual, the transverse cone of Δ along F). In particular $\ell(F^\circ) = 1$ if and only if the transverse cone is smooth.*

X is isolated-singular exactly when Δ has unit edges, and X° exactly when every two-dimensional face of Δ has $\ell(F^\circ) = 1$; a face with $\ell(F^\circ) = m \geq 2$ contributes to X° a curve of transverse A_{m-1} singularities (Proposition 4.1(2)). We obtain:

Corollary 1.2. *X and X° are both isolated-singular if and only if Δ has unit edges and every two-dimensional face of Δ is transversally smooth. We call such Δ* both-sides unit*. The condition is self-dual: Δ is both-sides unit if and only if Δ° is.*

Our main results determine the both-sides unit polytopes, and the local models they carry, *completely*, subject to one explicit input assumption. We use the following *database hypothesis*: the per-vertex-count files described in §5 contain, without repetition, exactly the members of the Kreuzer–Skarke classification in their stated vertex range; the unique remaining 36-vertex member is included separately. Under this hypothesis the scan covers all 473,800,776 reflexive 4-polytopes [KS00]. Every polytope returned by the scan is then re-verified in exact arithmetic, so the classification statements below have finite computer-assisted proofs conditional only on that database hypothesis. The transverse identity and the assertions about each explicitly displayed polytope are unconditional.

Theorem 1.3 (classification of both-sides local models)**.** *Under the database hypothesis above, there are exactly* 590 *both-sides unit reflexive* 4*-polytopes. Every two-dimensional face of every one of them is*

- *a triangle (the germ is a cyclic quotient point, or smooth), or*
- *a zonotope (a centrally-symmetric unit-edge polygon, always of type* (S)*), or*
- *a reflexive polygon: one of the five unit-edge polygons with exactly one interior lattice point, i.e. the Fano polygons of the five smooth toric del Pezzo surfaces.*

All five reflexive polygons occur. The complete face inventory is given in Table 2.

Theorem 1.4 (the non-smoothable germs of doubly isolated pairs)**.** *Under the database hypothesis, exactly three of the* 590 *both-sides unit polytopes carry a non-smoothable two-dimensional face. Writing $[k, i]$ for a unit-edge polygon with k edges and i interior lattice points, the germs occurring are:*

- $\frac{1}{3}(1,1,1)$ *(the cone over the* $[3,1]$ *triangle), six times on one* 9*-vertex polytope;*

- $\frac{1}{5}(1,1,3)$ *(the cone over the* $[3,2]$ *triangle), ten times on a single simplex;*
- *the anticanonical cone over* $\mathbb{F}_1$*, exactly* once*, on a unique* 22*-vertex polytope.*

In particular no deformable-but-non-smoothable (D) *face occurs on any both-sides unit polytope.*

Corollary 1.5. *Assume the database hypothesis. Let X be a unit-edge Batyrev Calabi–Yau threefold with a singular point of type* (D)*, for instance the pentagon-cone threefolds of* [Wue26a, Thms. 5.3, 5.8]*. Then the Batyrev mirror X° is non-isolated: it carries a curve of transverse A_{m-1} singularities for some $m \geq 2$. The deformable-but-non-smoothable phenomenon never occurs on a doubly isolated mirror pair.*

The one non-quotient exception is the most interesting object of this note.

Theorem 1.6 (the unique $\mathbb{F}_1$ mirror pair)**.** *Under the database hypothesis, up to lattice isomorphism there is exactly one doubly isolated Batyrev mirror pair whose singularities are not all quotient points or smoothable: the pair $(\Delta_{\mathbb{F}_1}, \Delta^\circ_{\mathbb{F}_1})$ of Theorem 6.1, with* 22 *and* 26 *vertices. The generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ has* 20 *ordinary double points, one* dP_6*-cone point, and one point whose germ is the anticanonical cone over* $\mathbb{F}_1$*; since that germ admits no smoothing at all, X* admits no smoothing [Wue26a, Cor. 4.3]*. Its mirror X° has* 26 *ordinary double points, two* dP_7*-cone points and two* dP_6*-cone points; every singular germ of X° is of type* (S)*. The maximal projective crepant partial resolutions have $(h^{1,1}, h^{2,1}) = (20, 26)$ and $(26, 20)$ respectively.*

Thus the Batyrev mirror of a Calabi–Yau threefold that is *frozen* (non-smoothable, carrying the cone over one of the two surfaces excluded in Gross's smoothing theorem for primitive contractions [Gro97, Thm. 5.8]) can be isolated-singular with every singular point locally smoothable. Notably, every dual edge on the mirror side has length one, so X° lies entirely in the $\ell(F^\circ) = 1$ regime of [Wue26b]. Its 26 nodes pose the *cross-face* problem of [Wue26b, §7]. Exact row reduction of their 26 diagonal-relation vectors gives rank 18 and two coloops, so the node subsystem has no relation with every coefficient non-zero (Remark 6.3). This would obstruct smoothing in an all-nodal threefold with a projective small resolution. The actual X°, however, also has four divisorial del Pezzo-cone points, so Friedman's nodal criterion does not apply to the mixed singularity configuration: their local deformation directions may participate in the global obstruction map. Its del Pezzo-cone germs are precisely those that [Wue26b, §6] places beyond Gross's theorems and for which no Friedman-type obstruction is known. Thus any smoothing would have to be genuinely mixed, and whether one exists remains open (Question 7.2).

Theorems 1.3, 1.4 and 1.6 answer the mirror question of [Wue26a, §6.3] for the non-smoothable local models: among them, only the cyclic quotients and the $\mathbb{F}_1$-cone, including the germs of Gross's two exceptional surfaces $\mathbb{P}^2$ and $\mathbb{F}_1$ [Gro97, Thm. 5.8], admit doubly isolated mirror realizations, and the $\mathbb{F}_1$-cone does so exactly once in the entire toric hypersurface landscape.

The opposite extreme of the classification carries a matching uniqueness. Exactly eight of the 590 polytopes have all two-dimensional faces smooth, so that X itself is smooth; for exactly one of them the mirror is smooth as well: the self-dual 24-cell, whose hypersurface is the classical self-mirror threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ [Bra12]. The smooth Batyrev mirror pair is thus unique in the entire classification (Example 5.4).

A caution about low-vertex evidence. An earlier version of this note conjectured that every two-dimensional face of a both-sides unit polytope is a triangle or a zonotope; equivalently, that no *asymmetric* face (one that is neither a triangle nor centrally symmetric) ever occurs. The conjecture is *true* for all 395,406,329 reflexive 4-polytopes with at most 18 vertices (83.5% of the classification) and *false* from 19 vertices on, exactly where both-sides unit polytopes first become common (Table 1): the dP_7 pentagon occurs (87 faces, on polytopes with 19 to 31 vertices), as does the $\mathbb{F}_1$ quadrilateral (once). See §5.3 for the refuted statements. Low-vertex evidence, however extensive, samples a biased corner of the landscape.

All mathematical predicates and reported invariants in the scan, the exact re-verification of all 590 polytopes, and the mirror-pair calculation use integer or rational arithmetic; the sole floating-point ordering operation is disclosed in §5.1. The programs build on those of [Wue26a] and are included as ancillary files; `both_sides_census.py` re-derives every number and theorem in this note from the scan results. The code and the scan data are public at `https://github.com/bwuebben/calabi-yau-smoothability`, where every count in this note is recomputable from `src/both_sides_census.py`.

Acknowledgements. This note completes the series [Wue26a, Wue26b]; the question of smoothing canonical Calabi–Yau singularities was posed to the author by Mark Gross many years ago.

2. Preliminaries

We keep the notation of [Wue26a]. For a lattice polygon Q with unit edges placed at height one, $C(Q)$ denotes the associated Gorenstein toric affine threefold; Q is of type (R), (D) or (S) according as its edge-vector multiset has no proper zero-sum sub-multiset, has one but no partition into unit segments and unimodular triples, or has such a partition [Alt97, CFP22]. Type (S) polygons are exactly those with smoothable cone; a single germ of type (R) or (D), one whose miniversal base has no smoothing component, obstructs every global smoothing of X [Wue26a, Lemma 4.1, Cor. 4.3]. We write $[k, i]$ for the class of a unit-edge polygon with k edges and i interior lattice points. A unit-edge polygon is a *zonotope* (a Minkowski sum of unit segments) if and only if it is centrally symmetric, and every zonotope is of type (S). A unit-edge polygon with exactly one interior lattice point is reflexive. Indeed, translate that point to the origin and let the polygon have k edges. Pick's theorem gives normalized area k, while the fan from the origin decomposes it into k lattice triangles of positive integral normalized areas, one over each unit edge. Every one therefore has normalized area one, so the origin has lattice distance one from every edge. There are five, the Fano polygons of $\mathbb{P}^2$, $\mathbb{F}_1$, $\mathbb{P}^1 \times \mathbb{P}^1$, dP_7, dP_6 (the subscript is the degree: $\mathrm{dP}_7 = \mathrm{Bl}_2\mathbb{P}^2$, $\mathrm{dP}_6 = \mathrm{Bl}_3\mathbb{P}^2$), the cone $C(Q)$ being the anticanonical cone over the corresponding surface. Of the five, the $\mathbb{P}^2$ triangle and the $\mathbb{F}_1$ quadrilateral are rigid and the other three are of type (S) [Wue26a, Rem. 2.6]. For a reflexive 4-polytope Δ we write $\Delta^\circ = \{u : \langle u, v\rangle \geq -1 \ \forall v \in \Delta\}$; a face $G \preceq \Delta$ has dual face $G^\circ \preceq \Delta^\circ$ with $\dim G + \dim G^\circ = 3$. By [Wue26a, Prop. 3.1], for Δ with unit edges the generic $X \in |-K|$ has, over each two-dimensional face F with $C(F)$ singular, exactly $\ell(F^\circ)$ singular points with germ $C(F)$, and no other singular points; and an edge of Δ of lattice length $m \geq 2$ produces a curve of transverse A_{m-1} singularities [Wue26a, Rem. 3.2]. Throughout, $\Delta \subset N_{\mathbb{R}}$ is the *fan* polytope, Batyrev's Δ^*, so our Δ° is his Δ; the same convention is used in [Wue26a, Wue26b].

3. The transverse-multiplicity identity

Let F be a codimension-two face of a reflexive polytope $\Delta \subset N_{\mathbb{R}}$ of arbitrary dimension, and let $u_1, u_2 \in M$ be the primitive inner normals of the two facets of Δ containing F, so that $\langle u_i, \cdot\rangle = -1$ on F and $F^\circ = \mathrm{conv}\{u_1, u_2\}$. [1] Put $M_F = M \cap (\mathbb{R}u_1 + \mathbb{R}u_2)$, a rank-two saturated sublattice of M containing u_1, u_2 as primitive vectors.

Lemma 3.1. $\ell(F^\circ) = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$.

Proof. Both sides depend only on the pair (u_1, u_2) inside the rank-two lattice M_F. Choose a $\mathbb{Z}$-basis of M_F in which $u_1 = (0, -1)$; this is possible because u_1 is primitive. Write $u_2 = (p, q)$; here $p \neq 0$, since $p = 0$ would force $u_2 = \pm u_1$, impossible as $F^\circ = \mathrm{conv}\{u_1, u_2\}$ is an edge of Δ° and 0 is interior to Δ°. Then $p = \det(u_1, u_2) = \pm[\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$, and $\ell(F^\circ) = \ell\big(\mathrm{conv}\{u_1, u_2\}\big) = \gcd(u_2 - u_1) = \gcd(p,\, q + 1)$, the two gcds agreeing because M_F is saturated in M, so lattice length in M may be computed in the chosen basis of M_F.

Let v be a vertex of F and let $\bar{v} \in N_F := N/(N \cap u_1^\perp \cap u_2^\perp)$ be its image; since u_1, u_2 vanish on $N \cap u_1^\perp \cap u_2^\perp$ they descend to elements of $M_F = N_F^\vee$, and $\bar{v}$ is a lattice point of $N_F \cong \mathbb{Z}^2$ with $\langle u_i, \bar{v}\rangle = -1$. Writing $\bar{v} = (a, b)$ in the basis of N_F dual to the chosen basis of M_F, $\langle u_1, \bar{v}\rangle = -b = -1$ gives $b = 1$, and $\langle u_2, \bar{v}\rangle = pa + q = -1$ gives $pa = -1 - q$, so $p \mid (q + 1)$. Hence $\gcd(p, q + 1) = |p|$, and $\ell(F^\circ) = |p| = [\, M_F : \mathbb{Z}u_1 + \mathbb{Z}u_2 \,]$. ☐

Remark 3.2. Lemma 3.1 is elementary and may be known; what matters here is the reading it provides. The proof works in every dimension for a codimension-two face. The right-hand side is the multiplicity of the two-dimensional cone $\mathrm{Cone}(u_1, u_2)$, which is the normal cone of F and dually the transverse cone of Δ along F (for two-dimensional cones the multiplicities of a cone and its dual agree). Thus *the dual-edge length counts exactly the transverse singularity multiplicity*, and $\ell(F^\circ) = 1$ is the statement that Δ is transversally smooth along F. (Transversal smoothness concerns only the directions normal to F: the germ $C(F)$ itself may well be singular.) The lattice-point hypothesis on v is essential: for an arbitrary pair of primitive vectors the two sides can differ.

Corollary 1.2 is immediate: X is isolated-singular iff Δ has unit edges; X° is isolated-singular iff Δ° has unit edges, i.e. every edge F° of Δ° has $\ell(F^\circ) = 1$, i.e. by Lemma 3.1 every two-dimensional face of Δ is transversally smooth. Self-duality follows since the conditions "unit edges" and "all dual edges of length one" are exchanged by $\Delta \leftrightarrow \Delta^\circ$.

4. The mirror correspondence of the singular strata

Proposition 4.1. *Let Δ be reflexive with unit edges and F any two-dimensional face. Under polar duality F corresponds to the edge F° of Δ°, and:*

1. *if $C(F)$ is singular, X has $\ell(F^\circ)$ isolated points of type $C(F)$ over F (and none over F otherwise);*
2. *X° has, along the surface $V(\sigma_{F^\circ}) \subset \mathbb{P}_{\Delta^\circ}$ (the orbit closure of the cone σ_{F° over the edge F° in the face fan of Δ°), a curve of transverse $A_{\ell(F^\circ)-1}$ singularities (and none if $\ell(F^\circ) = 1$).*

In particular X° is isolated-singular if and only if $\max_F \ell(F^\circ) = 1$, i.e. Δ is both-sides unit.

Proof. (1) is [Wue26a, Prop. 3.1]. For (2), the edge F° of Δ° has lattice length $\ell(F^\circ)$; an edge of lattice length m produces on the generic anticanonical hypersurface a curve of transverse A_{m-1} singularities (§2), and none when $m = 1$. Hence X° is isolated-singular iff every two-dimensional face of Δ has $\ell(F^\circ) = 1$. ☐

[1] The u_i are vertices of Δ°; [Wue26a] and the ancillary programs list the sign-flipped functionals taking the value $+1$ on the facet. Lattice lengths are unaffected.

Thus the $\ell(F^\circ)$ singular points of X over F and the $A_{\ell(F^\circ)-1}$-curve of X° correspond under polar duality, and the two threefolds are simultaneously isolated-singular exactly on the both-sides unit polytopes. Whenever Δ has *any* face with $\ell(F^\circ) \geq 2$ (the generic situation), the mirror is non-isolated and its singularities leave the isolated framework of [Alt97, CFP22]. Filip [Fil26] constructs, from a Laurent polynomial, a distinguished formal deformation of the associated non-isolated Gorenstein toric pair and proves that its general fibre is smooth exactly when the polynomial is 0-mutable; this does not classify all components of every non-isolated germ.

5. Enumeration over the Kreuzer–Skarke classification

5.1. **Methodology.** The both-sides unit condition is a condition on faces: all edges of lattice length one and, by Lemma 3.1, all dual edges of length one. We tested it over the per-vertex-count copy of the Kreuzer–Skarke classification used in [Wue26a, §6.1], with the production implementation built for that scan. Subject to the database hypothesis in the introduction, this covers the complete classification. All facet incidences, lattice lengths, determinants, and final predicates are checked in exact integer arithmetic, with no heuristic pruning. The implementation uses floating-point `atan2` only to choose a cyclic ordering of the vertices of an already identified planar face; the ensuing edge and lattice tests are integral. Implementation correctness is controlled in three complementary ways. (i) Before scanning each file of the database, the implementation was re-checked against an independently written reference implementation on random samples from that file. (ii) The 3,905,789 polytopes with at most nine vertices were scanned in full by both implementations, with identical results. (iii) The 590 polytopes found pass a global consistency check under polar duality: the both-sides condition is self-dual (Corollary 1.2), and the polar of every polytope found, computed exactly, is again both-sides unit and agrees with one of the 590 in vertex count, facet count and complete face inventory, an invariance that would generically fail for a false negative whose polar partner was detected. An invariant collision (another member of the 590 sharing its vertex count, facet count and face inventory) could in principle mask a miss, so this control is supporting evidence beside (i) and (ii), not an independent proof. The one polytope missing from the copy of the database we used, which is organized by vertex count, is the 36-vertex hexagon product (see [Wue26a, §6.1]); it is included. Every polytope passing the scan is then re-verified, and its faces classified, by the reference implementation; the ancillary scripts reproduce the full scan from the public database. The scan results are included as ancillary data files, and the program `both_sides_census.py` re-derives every statement of Theorems 1.3, 1.4 and 1.6 from them.

vertices	5	6	8	9	10	12	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	33	36
#	2	1	1	1	1	1	1	4	4	7	12	24	37	69	84	98	85	64	42	24	13	9	4	1	1

TABLE 1. The 590 both-sides unit reflexive 4-polytopes by vertex count (no others occur; in particular none with 7, 11, 13, 14, 32, 34 or 35 vertices). The population is closed under polar duality (Corollary 1.2) and peaks at 24 vertices.

5.2. **The classification.**

face $[k, i]$	kind	count	germ of X over it
$[3, 0]$	triangle	25,873	smooth point
$[3, 1]$	triangle	6	$\frac{1}{3}(1, 1, 1)$ **(rigid)**
$[3, 2]$	triangle	10	$\frac{1}{5}(1, 1, 3)$ **(rigid)**
$[4, 0]$	zonotope	15,652	ordinary double point
$[4, 1]$	zonotope	99	cone over $\mathbb{P}^1 \times \mathbb{P}^1$
$[4, 1]$	*asymmetric*	1	cone over $\mathbb{F}_1$ **(rigid)**
$[4, 2]$	zonotope	9	type (S)
$[5, 1]$	*asymmetric*	87	cone over dP_7
$[6, 1]$	zonotope	395	cone over dP_6, 2 smoothing components
$[6, 2]$	zonotope	2	type (S)
$[6, 7]$	zonotope	3	type (S)
$[8, 4]$	zonotope	1	type (S), 3 smoothing components

Table 2. Every two-dimensional face of every both-sides unit polytope, by class (k = edges, i = interior points). "Asymmetric" = neither a triangle nor centrally symmetric; only the two non-centrally-symmetric reflexive polygons occur, and each germ appears on X exactly once per face since all dual edges have length one. Type-(S) germs have one smoothing component except where noted.

Proof of Theorems 1.3 and 1.4. Both theorems are exhaustive verifications, summarized in Tables 1 and 2; the completeness of the underlying enumeration is the content of §5.1. Granting it, the scan of all 473,800,776 polytopes yielded 590 both-sides unit ones; exact re-analysis of each classifies all their two-dimensional faces into the twelve classes of Table 2, each of which is a triangle, a zonotope, or one of the five reflexive polygons (the $[4, 1]$ zonotope is the $\mathbb{P}^1 \times \mathbb{P}^1$ diamond, the $[6, 1]$ zonotope the dP_6 hexagon, the $[3, 1]$ triangle the $\mathbb{P}^2$ triangle; the asymmetric $[5, 1]$ and $[4, 1]$ classes are the dP_7 pentagon and the $\mathbb{F}_1$ quadrilateral, identified by explicit $GL_2(\mathbb{Z})$-equivalences). The non-smoothable classes occurring are the rigid triangles $[3, 1]$, $[3, 2]$ and the $\mathbb{F}_1$ quadrilateral; the $[3, 2]$ faces are the ten faces of the quotient simplex of Example 5.2, the $[3, 1]$ faces lie on the 9-vertex polytope of Example 5.2, and the $\mathbb{F}_1$ face on the unique polytope of Theorem 6.1. No face of type (D) occurs. □

Proof of Corollary 1.5. A type-(D) point of X is the cone over a type-(D) face F. By Theorem 1.4, Δ is then not both-sides unit, so some face F' has $\ell(F'^{\circ}) \geq 2$ and Proposition 4.1(2) gives a singular curve on X°. □

5.3. **The refuted conjectures.** The first version of this note conjectured that every two-dimensional face of a both-sides unit polytope is a triangle or a zonotope, and reduced this to a local "balancing" claim: an asymmetric face F with $\ell(F^{\circ}) = 1$, all faces of whose star have unit edges, forces some face of its star to have $\ell \geq 2$. The local claim had survived a complete check of the 38,571 candidate configurations arising among the 1,037,834 polytopes with at most eight vertices. Table 2 nevertheless refutes both statements: 88 asymmetric faces occur, on 55 polytopes. The smallest counterexample has 19 vertices:

$$\begin{gathered}\Delta_{19} = \mathrm{conv}\{(1,0,0,0),\ (0,1,0,0),\ (0,0,1,0),\ (0,0,-1,0),\ (0,-1,0,0),\\ (1,-1,-1,0),\ (0,0,0,1),\ (-1,1,1,-1),\ (0,-1,0,1),\ (0,0,-1,1),\\ (-1,1,0,-1),\ (-1,0,1,-1),\ (-1,0,1,0),\ (-1,1,0,0),\ (0,-1,-1,0),\\ (0,-1,-1,1),\ (-1,0,0,-1),\ (-2,1,1,-1),\ (-1,0,0,1)\},\end{gathered}$$

a both-sides unit polytope one of whose faces is the dP_7 pentagon; its hypersurface is a Calabi–Yau threefold with 14 nodes and one dP_7-cone point, all with isolated-singular mirror. That face is

$$P = \mathrm{conv}\big\{(0,-1,-1,0),\ (0,-1,-1,1),\ (-1,0,0,-1),\ (-2,1,1,-1),\ (-1,0,0,1)\big\},$$

the fifteenth to nineteenth entries of the list, lying in the two facets with inner normals $u_1 = (1,0,1,0)$ and $u_2 = (1,1,0,0)$; their difference $(0,-1,1,0)$ is primitive, so $\ell(P^\circ) = 1$. In the induced lattice its edge multiset is $\{(0,-1), (1,-1), (1,0), (-1,2), (-1,0)\}$, carried by $M = \left(\begin{smallmatrix}-1 & 0\\ 1 & 1\end{smallmatrix}\right) \in GL_2(\mathbb{Z})$ onto $\{(0,-1), (1,-1), (1,1), (-1,1), (-1,0)\}$, the dP_7 pentagon of [Wue26a, Rem. 2.6]: the unique unit-edge class with five vertices and one interior lattice point. The conjecture is *true* on the 395,406,329 polytopes with at most 18 vertices and false from 19 on — precisely where both-sides unit polytopes stop being sporadic (Table 1). We let the corrected statement stand as Theorem 1.3, and the conceptual question it raises as Question 7.1.

Remark 5.1 (why not "simplicial"). The stronger guess that a both-sides unit polytope is simplicial fails for a structural reason: the condition is self-dual, and "simplicial and simple" forces a simplex, contradicting the existence of both-sides unit polytopes with more than five vertices. The non-triangular faces are genuine; the content of Theorem 1.3 is that they are zonotopes or reflexive.

Example 5.2 (the low-vertex stratum)**.** Among the 3,905,789 polytopes with at most nine vertices exactly five are both-sides unit; four of them form two polar pairs. One pair is the smooth simplex and the simplex

$$\mathrm{conv}\{(1,0,0,0),\ (0,1,0,0),\ (2,4,5,0),\ (3,3,0,5),\ (-6,-8,-5,-5)\},$$

whose ten faces are the $[3,2]$ rigid triangles of Table 2 (quotient points of type $\frac{1}{5}(1,1,3)$) and whose polar is smooth. A second pair consists of a simplicial six-vertex polytope with all faces smooth and its nine-vertex polar, carrying the six $\frac{1}{3}(1,1,1)$ triangles and nine $[4,2]$ zonotopes. In each pair, the non-smoothable member mirrors a *smooth* threefold. The fifth polytope has eight vertices and sixteen facets, and all thirty-two of its faces are smooth triangles, so its hypersurface X is smooth. It shows that closedness under polar duality (Corollary 1.2) does not respect bounds on the number of vertices: its polar is a sixteen-vertex both-sides unit polytope all twenty-four of whose faces are $[4,1]$ diamonds, so the mirror X° carries twenty-four cone-over-$\mathbb{P}^1\times\mathbb{P}^1$ points.

Example 5.3 (the top of the classification)**.** The 36-vertex product $H \times H$ of two reflexive hexagons is the reflexive 4-polytope with the most vertices and the one polytope absent from the copy of the Kreuzer–Skarke database used in [Wue26a]. It is both-sides unit; its faces are 36 unit squares and 12 dP_6 hexagons, so its hypersurface has 48 singular points, all of type (S), while its polar, the 12-vertex free sum $H^\circ \oplus H^\circ$, has 72 smooth triangular faces and smooth hypersurface.

Example 5.4 (the unique smooth mirror pair)**.** Exactly eight of the 590 polytopes have every two-dimensional face smooth, so that X itself is smooth. For exactly one of them is the polar of the same kind: the 24-cell, self-dual, with 24 octahedral facets, 96 unimodular triangular faces, and no lattice points besides the origin and its 24 vertices. Its generic anticanonical hypersurface is the classical self-mirror Calabi–Yau threefold with $(h^{1,1}, h^{2,1}) = (20, 20)$ and $\chi = 0$; the $(1, 1)$-threefolds of Braun [Bra12] are free quotients of special symmetric members of this family. The scan shows the uniqueness: among all 473,800,776 reflexive 4-polytopes, the 24-cell is the only one whose generic anticanonical hypersurface and mirror are *both* smooth.

6. The unique $\mathbb{F}_1$ mirror pair

Theorem 6.1. *Let*

$$\begin{aligned}\Delta_{\mathbb{F}_1} = \operatorname{conv}\{&(1,0,0,0),\ (0,1,0,0),\ (1,-1,0,0),\ (0,0,1,0),\ (0,0,0,1),\ (0,0,1,-1),\\ &(0,0,-1,1),\ (0,0,0,-1),\ (0,0,-1,0),\ (-1,1,0,0),\ (0,-1,0,0),\ (-1,1,-1,1),\\ &(0,-1,0,-1),\ (0,-1,-1,0),\ (-1,1,-1,0),\ (-1,0,0,-1),\ (-1,0,-1,1),\\ &(-1,-1,0,-1),\ (-2,1,-1,1),\ (-2,1,-1,0),\ (-1,-1,-1,0),\ (-2,0,-1,0)\} \subset \mathbb{Z}^4.\end{aligned}$$

Then $\Delta_{\mathbb{F}_1}$ is a both-sides unit reflexive polytope with 22 vertices and 26 facets, and:

1. *its 72 two-dimensional faces are 50 smooth triangles, 20 unit squares, one* dP_6 *hexagon, and one quadrilateral $GL_2(\mathbb{Z})$-equivalent to the $\mathbb{F}_1$ polygon; every dual edge has lattice length one;*
2. *the generic anticanonical $X \subset \mathbb{P}_{\Delta_{\mathbb{F}_1}}$ is a Calabi–Yau threefold whose singular locus consists of* 20 *ordinary double points, one* dP_6*-cone point, and one point with germ the anticanonical cone over $\mathbb{F}_1$; since one germ without a smoothing component obstructs every global smoothing (§2), X admits no smoothing* [Wue26a, Cor. 4.3]*;*
3. *the polar $\Delta^\circ_{\mathbb{F}_1}$ (26 vertices, 22 facets) is both-sides unit with* 68 *two-dimensional faces:* 38 *smooth triangles,* 26 *unit squares, two* dP_7 *pentagons and two* dP_6 *hexagons; the mirror X° has* 26 *ordinary double points, two* dP_7*-cone points and two* dP_6*-cone points, each with a local smoothing;*
4. *the maximal projective crepant partial resolutions satisfy* $(h^{1,1}, h^{2,1})(\widehat{X}) = (20, 26)$ *and* $(h^{1,1}, h^{2,1})(\widehat{X^\circ}) = (26, 20)$*;*
5. *under the database hypothesis, $\Delta_{\mathbb{F}_1}$ is the* only *both-sides unit reflexive* 4*-polytope, up to lattice isomorphism, carrying a non-smoothable face that is not a triangle.*

Proof. (1)–(4) are finite checks on the explicitly listed polytope, a face enumeration reproducible in any lattice-polytope package, carried out in exact arithmetic by the ancillary files.[2] We record the $\mathbb{F}_1$ face itself, since it is the object the statement is about. Its four vertices are the fourth, eighteenth, nineteenth and twenty-second entries of the list above,

$$F = \operatorname{conv}\big\{(0,0,1,0),\ (-1,-1,0,-1),\ (-2,1,-1,1),\ (-2,0,-1,0)\big\},$$

and the two facets of $\Delta_{\mathbb{F}_1}$ containing F have inner normals

$$u_1 = (1,1,-1,-1), \qquad u_2 = (1,0,-1,0),$$

in the normalization of §1: both take the value -1 at each of the four vertices of F and are ≥ -1 on $\Delta_{\mathbb{F}_1}$. Their difference $u_1 - u_2 = (0,1,0,-1)$ is primitive, so $\ell(F^\circ) = 1$. In the

[2] Ancillary files `both_sides_census.py` and `face_data.py`.

induced lattice on the affine span of F the edge multiset is $\{(0,-1),\ (1,-1),\ (1,1),\ (-2,1)\}$, and

$$M = \begin{pmatrix} -1 & -1 \\ -1 & 0 \end{pmatrix} \in GL_2(\mathbb{Z}), \qquad \det M = -1,$$

carries it onto $E(Q_A) = \{(-2,-1),\ (0,-1),\ (1,0),\ (1,2)\}$, the edge multiset of the $\mathbb{F}_1$ polygon of [Wue26a, Rem. 2.6]; so $C(F)$ is the anticanonical cone over $\mathbb{F}_1$.

The dP_6 hexagon of (1) is recorded the same way. It is the face

$$G = \operatorname{conv}\big\{(0,0,-1,0),\ (0,-1,-1,0),\ (-1,1,-1,0),\\ (-2,1,-1,0),\ (-1,-1,-1,0),\ (-2,0,-1,0)\big\}$$

(the ninth, fourteenth, fifteenth, twentieth, twenty-first and twenty-second entries of the list), lying in the two facets with inner normals $u_1 = (0,0,1,0)$ and $u_2 = (0,0,1,1)$, whose difference $(0,0,0,-1)$ is primitive, so $\ell(G^\circ) = 1$. Its edge multiset in the induced lattice is already the standard hexagon,

$$E(G) = \{(0,-1),\ (1,-1),\ (1,0),\ (0,1),\ (-1,1),\ (-1,0)\},$$

so no change of basis is needed: $C(G)$ is the anticanonical cone over dP_6, the class with two smoothing components in the catalogue of [Wue26a, Rem. 2.6]. The Hodge numbers are given by Batyrev's formulas [Bat94, §4], evaluated in exact arithmetic. The mirror swap in (4) is forced by Batyrev duality and serves as a consistency check. Conditional on the database hypothesis, (5) is Theorem 1.4: the scan finds no other non-triangle non-smoothable face on any both-sides unit polytope. □

Remark 6.2 (a threefold with no smoothing whose mirror has only locally smoothable singularities). Theorem 6.1 pairs a Calabi–Yau threefold that cannot be smoothed (its $\mathbb{F}_1$-cone point is rigid, with reduced miniversal base a point, the local model of Gross's exceptional case) with a mirror all of whose singular points move locally. Whether these local deformations assemble to a global smoothing of X° is exactly the frontier of [Wue26b]: all $26+2+2$ of its singular points sit over faces with $\ell(F^\circ) = 1$, so the single-face smoothing techniques of [Wue26b] do not apply. The node subsystem fails the all-nonzero relation of the nodal problem (Remark 6.3), while the four del Pezzo-cone points are the $\ell = 1$ germs that [Wue26b, Prop. 6.1, Rem. 6.2] place beyond Gross's theorems, with no Friedman-type obstruction known. The mixed local-to-global problem remains open. The pair thus concentrates, on two mirror-dual threefolds, every open smoothability phenomenon of this series.

Remark 6.3 (the node subsystem). Let $q_1, \dots, q_{26}$ be the vertices of $\Delta^\circ_{\mathbb{F}_1}$. For each of its 26 unit-square faces $S = (q_a, q_b, q_c, q_d)$ in cyclic order, form the diagonal-relation vector

$$\rho_S = e_a - e_b + e_c - e_d \in \mathbb{Q}^{\mathrm{Vert}(\Delta^\circ_{\mathbb{F}_1})}.$$

Exact row reduction of the resulting 26×26 matrix gives rank 18. Exactly two rows are coloops. The corresponding square faces, displayed in cyclic order, are

$$S_- = \operatorname{conv}\{(-1,-1,0,-1),\ (-1,0,0,-1),\ (0,1,0,-1),\ (0,0,0,-1)\},$$
$$S_+ = \operatorname{conv}\{(-1,-1,0,1),\ (0,-1,0,1),\ (0,0,0,1),\ (-1,0,0,1)\}.$$

Deleting either row lowers the rank to 17, deleting both lowers it to 16, and deleting any other row leaves rank 18. Thus every linear dependence among the 26 node rows has coefficient zero on both ρ_{S_-} and ρ_{S_+}; in particular there is no dependence with all coefficients non-zero. The calculation is exact and is reproduced by the ancillary row-reduction script. [3]

[3] `paper3_node_relations.py`.

For a hypothetical all-nodal threefold this fails the Friedman–Batyrev–Kreuzer condition recalled in [Wue26b, §7]. It does *not* decide the actual X°: its four additional del Pezzo-cone points have divisorial crepant resolutions, and their local smoothing directions may contribute to the mixed local-to-global obstruction map. What the computation proves is that a purely nodal cross-face rescue is impossible; any smoothing of X° must couple the node directions to the del Pezzo directions.

7. Questions

Question 7.1. Why triangles, zonotopes, and reflexive polygons? Theorem 1.3 is an exhaustive fact about dimension four; a conceptual, dimension-free proof (a local characterization of the unit-edge polygons admitting an all-unit, all-transversally-smooth reflexive completion) would explain it. The failed local balancing statement of §5.3 shows the answer is subtler than the star of the face; the appearance of exactly the *reflexive* polygons as the non-zonotope exceptions is suggestive in a polar-duality-symmetric problem.

Question 7.2. Is the mirror X° of Theorem 6.1 smoothable? All its germs are of type (S) with $\ell(F^\circ) = 1$. A positive answer would exhibit a Batyrev mirror pair with one member rigid against smoothing and the other smoothable — a maximally asymmetric pair for the deformation–resolution exchange of mirror symmetry. The node subsystem alone admits no all-nonzero relation (Remark 6.3); can the four del Pezzo deformation directions supply the required mixed local-to-global cancellation?

Question 7.3. Corti–Filip–Petracci conjecturally relate smoothing components of $C(Q)$ to 0-mutable Laurent polynomials with Newton polygon Q [CFP22], and Filip [Fil26] constructs distinguished formal deformations in the non-isolated case, detecting when their general fibres are smooth. In that language, what distinguishes the twelve face classes of Table 2 — and what is the mirror-theoretic meaning of the unique $\mathbb{F}_1$ face? One would like to read Theorem 1.4 off the canonical scattering diagram of the mirror.

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